

# SEBASTIAN CARABALLO

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## EDUCATION

### Purdue University

B.S. in Mechanical Engineering, Minor in Computer Science

West Lafayette, IN

Exp. June 2028

- **Honors:** Dean's List and Semester Honors ('24), Alpha Lambda Delta Phi Eta Sigma Honors Society
- **Coursework:** Linear Algebra, Data Science, Multivariable Calculus, Statistics, Dynamics, Diff. Equations

## SKILLS

- **Languages:** Python, JavaScript/TypeScript (React, Node.js), SQL, MATLAB, C/C++, Java, C#, HTML/CSS, Arduino
- **Software:** Next.js, FastAPI, PostgreSQL, Git, Vercel, JIRA, AutoCAD, SolidWorks, Siemens NX, FEA, MeshMixer
- **Equipment:** Wiring, Soldering, DAC/ADC boards, Laser Cutting, 3D Printing, CNC, GD&T, Raspberry Pi, Sensors

## EXPERIENCE

### FitScript | [fitscript.io](https://fitscript.io)

Founder / CEO

West Lafayette, IN

May 2026 - Present

- Founded FitScript, an AI nutrition coach for diabetic and GLP-1 users, and built the full-stack web app solo in Next.js, TypeScript, and PostgreSQL. Launched a private beta and onboarded 60+ users.
- Selected a diabetic-first market with a custom segment-scoring tool that ranked 3 customer segments against a locked 6-criteria rubric. Track retention through SQL analytics on meal and chat data to guide the roadmap.

### Algoma

Software Development Engineer Intern

Brooklyn, NY

May 2025 - Aug 2025, June 2026 - Present

- Returned for a second summer in a deeper engineering role. Ship production features and UX fixes to the React property-discovery platform through Jira-ticketed pull requests and code review.
- Designed and programmed interactive models for a client-facing analytics page, enabling analysis of up to 10 filtered properties from hundreds. Client feedback in 8+ demos highlighted the improved interface.
- Implemented custom data validation and weighting algorithms to analyze complex variables, cutting production risk and improving system performance to help secure 20% of clients by Q3.

### Vlachos Research Group | Cardio Fabrication & Flow Loop Teams

Research Assistant

West Lafayette, IN

January 2026 - Present

- Led system bring-up of a fluidic testbed, interfacing DAC/ADC boards with high-fidelity sensors. Wrote automated Python workflows that synchronize high-speed imaging with real-time sensor data.
- Designed and fabricated precision casting molds and 0.3x-scale models with additive manufacturing. Ran producibility reviews to find bottlenecks and turn models into manufacturable hardware.

### Purdue National Security & Defense Society (NSDS)

R&D Member, Unmanned Systems

West Lafayette, IN

December 2025 - Present

- Developing a rapid-deployment RF payload system, creating detailed mechanical Tooling Build-to-Packages (BTP) and engineering drawings to meet strict structural durability requirements.
- Designing modular flight-controller mounting interfaces in AutoCAD and aligning firmware compatibility across microcontrollers and ESCs for stable quadcopter payload deployment.

### PURPL (Purdue Undergraduate Rocket Propulsion Lab)

Turbojet Sub-Team Member

West Lafayette, IN

August 2024 - Present

- Architected 3D layouts and arrangements for a 50-lbf turbojet, ensuring structural integrity of hardware components for high-pressure fuel delivery and space-relevant mechanical interfaces.
- Conducted high-speed core testing at 80,000 RPM and ran Finite Element Analysis (FEA) trade studies to evaluate hardware performance under extreme thermal and mechanical loads.

## PROJECTS

### Earnings Trader | [Live Demo](#)

Personal Project | LLM Trading-Signal and Backtesting Engine

Python, FastAPI, React

May 2026 - Present

- Built an LLM pipeline that reads SEC EDGAR 8-K filings and scores earnings language for sentiment, guidance quality, and risk with GPT-4o-mini. Generated 50+ trade signals across 10 tickers.
- Engineered a vectorbt engine backtesting 35+ trades against an SPY benchmark on Sharpe, win rate, and drawdown. The system runs itself on scheduled CI, ingesting new filings nightly and redeploying a live FastAPI and React dashboard.

## AWARDS/PUBLICATIONS

- **Publications:** Skomra A., Russon A., Chevli K.K., Caraballo S. "Combination of Pre-Biopsy ExoDx Testing and mpMRI Improves Predictive Power of High-Grade Prostate Cancer." Journal of Urology, 2023.
- **Awards:** FRC Regional Awards: Industrial Design ('22), Creativity ('23), Engineering Inspiration ('23)
- **Involvement:** American Society of Mechanical Engineers (ASME), Undergraduate Teaching Assistant, ΘX Fraternity